AI Image Generator

Technical Documentation, Workflows and Requirements

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Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

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1. Project Overview

1.1 Purpose

AI Image Generator is a self-hosted application that turns a plain-English prompt into an image using Stable Diffusion XL, its best community fine-tunes, or FLUX.1, running entirely on a local NVIDIA GPU. No image data leaves the machine and there is no per-image cost.

The project was built on 21 and 22 October 2026 and grew through eight increments:

1. A command-line script that runs Stable Diffusion XL on the GPU.
2. A local web interface so a prompt can be typed and the result viewed in a browser.
3. A prompt language that replaces sliders and dropdowns: size, ratio, quality, style, count and seed are all read from the prompt text.
4. Per-image telemetry (timings, VRAM, temperature, power, clocks) shown in the UI and stored in a SQLite database with a History tab.
5. A Real-ESRGAN upscaling stage so that "4k" or "8k" in the prompt produces a genuine 3840 or 7680 pixel image.
6. A model registry: six SDXL-family checkpoints, Z-Image Turbo and FLUX.1 schnell, automatic model choice by style, local checkpoints, LoRAs, prompt weighting and per-model samplers.
7. The Frazile AI studio interface: dark theme with pink-to-yellow accents, full-width layout, and an Upscale bar that appears after generation.
8. An Admin panel with live CPU, GPU, memory, storage, network and power telemetry, plus a Docker image for container deployment.

1.2 Key capabilities

- Text-to-image generation with eight built-in models: Z-Image Turbo (the default), SDXL base 1.0, Juggernaut XL v9, RealVisXL V5.0, DreamShaper XL v2 Turbo, Playground v2.5, Animagine XL 3.1 and FLUX.1 schnell, all in 16-bit precision (FLUX in 4-bit on 24 GB cards).
- Any SDXL checkpoint dropped into models/checkpoints/ and any SDXL LoRA in models/loras/ is usable immediately after a restart.
- Prompt-driven control of model, LoRAs, emphasis weighting, aspect ratio, exact resolution, orientation, quality, fidelity, art style, image count, seed, steps and guidance.
- Automatic model selection from style words (photorealistic goes to Juggernaut, anime to Animagine, painting to Playground, digital art to DreamShaper).
- Live interpretation line under the prompt box showing exactly what will be generated, with which model, before the button is pressed.
- Tiled Real-ESRGAN upscaling to 2K, 4K, 8K or any exact size, either from the prompt or afterwards from the Upscale bar, with automatic centre crop.
- Full telemetry per image: inference time, steps per second, peak VRAM, GPU temperature and power before and after, clocks, fan, performance state, driver and library versions.
- Persistent history in SQLite with search, aggregate statistics, and click-to-view of any past image with its full record, including later upscales.
- Admin panel: nine live tiles and six detail sections covering GPU, CPU, memory and storage, network, power, software and the app process, with optional 5-second auto-refresh.
- Command-line tools for batch generation with any model and LoRA, upscaling an existing image, querying the database and dumping system telemetry.

- A Dockerfile and docker-compose.yml for running the whole studio as a GPU container.

1.3 Target environment

The application was developed and verified on the following machine. Other NVIDIA GPUs with at least 12 GB of VRAM will work for the SDXL-family models; see the Requirements section for the full matrix.

Item	Value
GPU	NVIDIA GeForce RTX 3090, 24 GB VRAM, compute capability 8.6, 82 SMs
CPU / RAM	AMD Ryzen 5 3500 (6 cores), 16 GB
Driver	610.88
OS	Windows 11 Pro 10.0.26300
Python	3.12.0
CUDA runtime (via PyTorch)	12.6
Project folder	C:\Users\frazi\Documents\apps\ai-image-gen
Repository	https://github.com/Fzhacking/ai-image-gen (private)

1.4 Document map

Section	Contents
2. Requirements	Functional and non-functional requirements, hardware and software matrix, model assets
3. Architecture	Components, pipeline diagram, data flow, design decisions, concurrency
4. Workflows	Setup, generation, choosing models, LoRAs, upscaling, history, Admin panel, Docker, maintenance
5. Module reference	Every source file, its public functions and their contracts
6. Models	The model registry, automatic selection, samplers, prompt weighting, LoRAs, FLUX
7. Prompt language	Complete reference of words the parser understands
8. Database	Schema of generations.db, every column explained, example queries
9. Admin panel	Every tile and section of the live system panel and where its numbers come from
10. Performance	Measured timings and resource use on the RTX 3090 for every model
11. Deployment and Docker	Container image, compose file, volumes, GPU access, host requirements
12. Operations	Running, stopping, troubleshooting, known issues, backup, security
13. Change log and roadmap	What was built in which order, defects fixed, future work
Appendix	File tree, dependency list, command cheat sheet, API access, licences

2. Requirements

2.1 Functional requirements

ID	Requirement	Status
FR-01	Generate an image from a text prompt using a locally hosted diffusion model on the GPU.	Done
FR-02	Provide a browser interface where a prompt can be typed and the result is displayed.	Done
FR-03	Keep the model loaded between requests so repeat generations take seconds, not minutes.	Done
FR-04	Control aspect ratio, resolution, orientation, quality, fidelity, style, count and seed from words in the prompt, with no separate controls.	Done
FR-05	Show how the prompt was interpreted before generation starts.	Done
FR-06	Produce genuine 4K (3840 px) and 8K (7680 px) output when requested, and exact sizes such as 3840x2160.	Done
FR-07	Display, for every image, the GPU used, time taken, VRAM, temperature, power, clocks and software versions.	Done
FR-08	Persist every generation with its settings and telemetry in a database that can be browsed later.	Done
FR-09	Provide a History view with search, aggregate statistics and click-to-view of past images.	Done
FR-10	Save every generated image, native and upscaled, as a PNG file on disk.	Done
FR-11	Support a negative prompt describing what to avoid.	Done
FR-12	Provide command-line equivalents for generation, upscaling and database queries.	Done
FR-13	Offer several models with different strengths and let the prompt or a dropdown choose between them; switch models without restarting.	Done (v2.0)
FR-14	Choose a suitable model automatically from the style words in the prompt.	Done (v2.0)
FR-15	Accept user-supplied SDXL checkpoints and LoRAs from local folders.	Done (v2.0)
FR-16	Support emphasis weighting such as (word:1.3) and prompts longer than 77 tokens.	Done (v2.0)
FR-17	Upscale an already generated image from the UI without regenerating it, recording the result on the same history row.	Done (v2.0)
FR-18	Present the application in Frazile AI branding with a full-width layout.	Done (v2.0)
FR-19	Provide an Admin panel with live CPU, GPU, memory, storage, network, power and software information.	Done (v2.0)
FR-20	Run the whole application as a GPU-enabled Docker container.	Done, untested on the development machine (no Docker installed)

2.2 Non-functional requirements

ID	Requirement	How it is met
NFR-01	Runs fully offline after the first model download.	Model weights cached in the Hugging Face cache; upscaler weights stored in models/.
NFR-02	A 1 megapixel image generates in under 15 seconds on an RTX 3090.	Measured 2.0 to 10.5 s depending on model, steps and size.
NFR-03	Peak VRAM stays under 16 GB so 16 GB cards can run every SDXL-family model.	Measured peak 13.2 GB at 832x1216 with prompt weighting; 10.5 GB at 1024x1024 without.
NFR-04	Upscaling must never exceed VRAM regardless of output size.	512 px tiles with 32 px overlap; memory is bounded per tile.

ID	Requirement	How it is met
NFR-05	Switching models must free the previous model's VRAM.	ModelManager drops all references, runs gc and empties the CUDA cache; measured 6.58 GB to 0.01 GB resident.
NFR-06	The server binds to localhost only unless deliberately configured otherwise.	Gradio launched on 127.0.0.1:7860; the Docker image binds 0.0.0.0 inside the container only.
NFR-07	All persistent state is in plain files that can be backed up by copying.	PNG files in outputs/, SQLite file generations.db, weights in models/ and the HF cache.
NFR-08	The database schema can evolve without losing data.	db.connect() adds missing columns automatically.
NFR-09	The Admin panel must not slow the GPU work.	A refresh costs about 0.7 s after the first; folder sizes and thread counts are cached for 60 and 30 s.
NFR-10	Code is dependency-light and readable: no framework beyond Gradio and diffusers.	Eight Python modules, about 2,500 lines in total.

2.3 Hardware requirements

Component	Minimum	Recommended	Verified on
GPU	NVIDIA, 12 GB VRAM, compute capability 7.0+ (SDXL family); 24 GB for Z-Image Turbo and FLUX	24 GB	RTX 3090, 24 GB
System RAM	16 GB	32 GB	16 GB
Disk	40 GB free (Z-Image Turbo 20 GB plus its 12 GB bf16 cache, 64 MB upscaler, Python environment, outputs)	120 GB: every SDXL model is 6.5 GB and FLUX is 33 GB	250 GB free, 71 GB used by models
CPU	Any x86-64	6 cores or more	Ryzen 5 3500

Cards below 12 GB can run the command-line script with CPU offload, but the web app assumes an SDXL pipeline fits in VRAM. FLUX on a card under 40 GB loads its transformer in 4-bit and its T5 encoder in 8-bit with per-model CPU offload.

2.4 Software requirements

Package	Version used	Purpose
Python	3.12.0	Runtime
torch	2.14.1+cu126	Tensor library with CUDA 12.6
torchvision	0.29.1+cu126	Required by transformers image processors and spandrel
diffusers	0.41.0	SDXL and FLUX pipelines, schedulers, LoRA loading
transformers	5.20.2	CLIP and T5 text encoders
accelerate	1.15.0	Device placement and CPU offload
safetensors	0.8.0	Safe weight loading
huggingface_hub	1.33.0	Model download and cache
peft	0.21.2	LoRA adapters on the UNet and text encoders
compel	2.5.1	Prompt weighting and long prompts for SDXL
bitsandbytes	0.50.2	4-bit and 8-bit quantisation for FLUX
sentencepiece, protobuf	0.2.2, 7.36.2	T5 tokenizer used by FLUX
gradio	6.30.0	Web interface, theme, timer
spandrel	0.4.2	Loads the Real-ESRGAN architecture and weights
pandas	3.0.6	History table in the UI
psutil	7.2.2	CPU, memory, disk and network statistics for the Admin panel

Package	Version used	Purpose
Pillow	12.3.0	Image I/O, resizing, cropping, figure assembly
numpy	2.5.4	Tile arrays for the upscaler
reportlab	5.0.1	Builds this documentation (development only)
NVIDIA driver	610.88	Also provides nvidia-smi for telemetry

2.5 Model assets

Asset	Source	Size	Location
stabilityai/stable-diffusion-xl-base-1.0 (fp16)	Hugging Face Hub	6.7 GB	HF cache
RunDiffusion/Juggernaut-XL-v9 (fp16)	Hugging Face Hub	6.5 GB	HF cache
SG161222/RealVisXL_V5.0 (fp16)	Hugging Face Hub	6.5 GB	HF cache
Lykon/dreamshaper-xl-v2-turbo (fp16)	Hugging Face Hub	6.5 GB	HF cache
playgroundai/playground-v2.5-1024px-aesthetic (fp16)	Hugging Face Hub	6.5 GB	HF cache
cagliostrolab/animagine-xl-3.1	Hugging Face Hub	6.5 GB	HF cache
Tongyi-MAI/Z-Image-Turbo (bf16)	Hugging Face Hub	20 GB	HF cache (+12 GB bf16 copy in models/cache/)
Qwen/Qwen-Image transformer, VAE, tokenizer	Hugging Face Hub	41 GB	HF cache (+ ~11 GB 4-bit copy in models/cache/)
diffusers/qwen-image-nf4 text encoder	Hugging Face Hub	5.8 GB	HF cache
lightx2v Qwen-Image-Lightning 8-step V2.0 LoRA	Hugging Face Hub	0.8 GB	HF cache
black-forest-labs/FLUX.1-schnell	Hugging Face Hub (gated, login required)	33 GB	HF cache, not yet downloaded
RealESRGAN_x4plus.pth	github.com/xinntao/Real-ESRGAN releases v0.1.0	64 MB	models/ inside the project
User checkpoints and LoRAs	Civitai or own training	varies	models/checkpoints/, models/loras/

The Hugging Face cache is `%USERPROFILE%\.cache\huggingface\hub` on Windows and the `hf-cache` volume inside Docker.

2.6 Constraints and assumptions

- SDXL-family models are trained at about 1 megapixel. Native generation is therefore capped at a 1536 px long side; larger sizes are reached by upscaling.
- Only one model is resident at a time. Switching takes about 26 s for an SDXL-family model and longer for Z-Image Turbo; the first use of a model downloads it.
- Z-Image Turbo and FLUX.1 schnell have no negative prompt and no guidance scale, and LoRAs and prompt weighting are not applied to them. FLUX also requires a one-time Hugging Face login because the repository is gated.
- The telemetry module depends on nvidia-smi being on the PATH, which the NVIDIA driver installer guarantees on Windows. If it is missing, telemetry fields are stored as NULL and generation still works.
- The application is single-user and single-GPU. Concurrent requests are queued by Gradio and processed one at a time.
- Each model's licence (see Appendix E) permits local and commercial use with its own use restrictions; the operator is responsible for the content they generate.

3. System Architecture

3.1 Component overview

The application is a single Python process. Gradio serves the browser interface and calls plain Python functions; those functions use seven project modules, one resident diffusion model and the upscaler.

Component	File	Responsibility
Web application	app.py	Builds the Gradio UI and theme, owns the ModelManager and the Upscaler, orchestrates one generation end to end, renders the details report, serves the History and Admin tabs.
Model registry	models.py	Registry of built-in and local models, ModelManager (load, swap, free VRAM), samplers, LoRA activation, compel prompt encoding, FLUX quantised loading.
Prompt parser	prompt_parser.py	Turns free text into a Settings object: model, LoRAs, size, steps, guidance, count, seed, style, upscale target. Pure function, no GPU.
Upscaler	upscaler.py	Wraps Real-ESRGAN through spandrel; tiled x4 passes; resizes or crops to the requested target.
GPU telemetry	gpu_info.py	Static GPU facts from torch; live temperature, power, clocks, fan and load from nvidia-smi.
System telemetry	sysinfo.py	CPU, memory, disks, network, GPU extras, power, software and process facts for the Admin panel, built on psutil and nvidia-smi.
Database	db.py	SQLite schema, migration, insert, update, fetch, search, per-row get, aggregate stats, command-line viewer.
CLI generator	generate.py	Standalone batch generation with any model and LoRA, without the web UI.

3.2 Generation pipeline

For every request the sequence is:

1. The prompt text is parsed into a Settings object. Model and LoRA tags and numeric directives are stripped; style words add descriptor suffixes.
2. The model is chosen: named in the prompt, selected in the Model box, or picked automatically from the style. The model's own step and guidance defaults are applied unless the prompt set them.
3. If the chosen model is not the resident one, the ModelManager frees the current pipeline and loads the new one with its recommended sampler.
4. Requested LoRAs are loaded (once) and activated with their weights; with none requested, adapters are disabled.
5. The prompt and negative prompt are encoded with compel into weighted embeddings (SDXL family) so that (word:1.3) and long prompts work; FLUX receives plain text.
6. For each image: the seed is fixed, peak-memory counters are reset and a telemetry snapshot is taken.
7. The pipeline runs for the requested number of steps at the native size. A per-step callback drives the progress bar.
8. CUDA is synchronised, the inference time is recorded, and a second telemetry snapshot is taken while the GPU is still under load.
9. The native image is saved as PNG. If an upscale target is set, Real-ESRGAN runs and the upscaled PNG is saved.
10. A record with about 50 fields is inserted into generations.db; the gallery, details report, Upscale bar and "Loaded now" line are updated in the browser.

A later click on the Upscale bar reads the native PNG of each selected record, upscales it, saves the result and writes the eight upscale columns onto the same row with db.update().

3.3 Data flow and storage

Data	Where it lives	Format	Written by
Native images	outputs/YYYYMMDD-HHMMSS_seedN.png	PNG, 1 to 2 MB	app.py, generate.py
Upscaled images	outputs/..._seedN_3840px.png	PNG, 8 to 31 MB	app.py
Generation records	generations.db	SQLite, one table	db.py
Built-in model weights	Hugging Face cache	safetensors	downloaded once
User checkpoints and LoRAs	models/checkpoints/, models/loras/	safetensors	the user
Upscaler weights	models/RealESRGAN_x4plus.pth	pth	downloaded once
Logs	stdout of the app process	text	app.py

3.4 Design decisions

Prompt as the only control surface. Early versions had dropdowns for style and size and sliders for steps and guidance. These were removed in favour of a parser because a single text box is faster to use, is self-documenting through the live interpretation line, and makes every generation reproducible from the prompt string alone. The Model box is the one exception, and even it can be overridden from the prompt.

One resident model, swapped on demand. Holding every model in VRAM would need 40 GB. The ModelManager keeps exactly one pipeline, swaps in about 26 s, and verifies the old one is freed. Automatic selection only switches to models that are already on disk so a style word never triggers a surprise 6.5 GB download.

Per-model defaults and samplers. A turbo model wants 6 steps at guidance 2, FLUX wants 4 steps at 0, SDXL wants 30 at 7. The parser records whether the prompt set steps or guidance explicitly; otherwise the model supplies them, and quality words scale them proportionally. Every SDXL-family model runs with the sampler its author recommends instead of the stock Euler scheduler.

compel for prompt weighting. The CLIP encoders truncate at 77 tokens and have no notion of emphasis. compel chunks long prompts and applies (word:1.3) weights to the embeddings. Its own padding helper does not support SDXL's dual encoder, so models.py pads mismatched chunk counts with the embedding of the empty prompt.

Two-stage resolution. SDXL produces incoherent output above roughly 1 megapixel. Generating at native size and upscaling with a super-resolution network is the standard approach and completes in a few seconds. The Upscale bar makes the second stage available after the fact, on the same database row.

Tiled upscaling. Real-ESRGAN on a full 3840 px image would need tens of gigabytes for intermediate feature maps. Processing 512 px tiles with a 32 px overlap bounds memory and hides seams.

Telemetry via nvidia-smi and psutil. Shelling out to nvidia-smi avoids a dependency on NVML bindings and costs about 100 ms per call. psutil covers CPU, memory, disks and network portably, so the Admin panel works unchanged inside the Docker image.

SQLite over a server database. A single-user desktop tool needs no server. The file is trivially backed up and any SQL client can open it. connect() adds missing columns so the schema can grow without migration scripts.

Always-dark theme. The Frazile AI palette is applied to both of Gradio's colour modes so the studio looks the same regardless of the browser's preference.

3.5 Concurrency model

Gradio runs the generate and upscale functions in a worker thread and queues additional requests. The GPU is used by one job at a time. The History and Admin tabs read the database and the system on their own threads; SQLite handles the concurrent reads safely because each call opens its own short-lived connection, and the Admin collectors never touch the pipeline.

4. Workflows

4.1 First-time setup

1. Confirm the GPU and driver are visible: run nvidia-smi in a terminal. It must list the card and a driver version.
2. Open PowerShell in the project folder.
3. Create and activate a virtual environment: `python -m venv .venv` then `.\.venv\Scripts\Activate.ps1`
4. Install the CUDA build of PyTorch, then the remaining dependencies: `pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126` and `pip install -r requirements.txt`
5. Verify CUDA is active: `python -c "import torch; print(torch.cuda.is_available(), torch.cuda.get_device_name(0))"`
6. Confirm the upscaler weights exist at models\RealESRGAN_x4plus.pth. If not, download them from the Real-ESRGAN v0.1.0 release page into that folder.
7. Start the app once. The first start downloads the 20 GB Z-Image Turbo model into the Hugging Face cache. Other models download the first time they are used.

4.2 Daily use: generating an image

1. Double-click run-app.bat, or run `.\.venv\Scripts\python.exe app.py` from the project folder.
2. Wait for "Z-Image Turbo ready" and "Upscaler ready" in the console (about a minute). The browser opens http://127.0.0.1:7860 automatically.
3. Type a prompt. As you type, the line under the box shows the interpreted settings and the model, for example "1344x768 | upscale to 3840x2194 | 40 steps | guidance 5 · model Juggernaut XL v9".
4. Optionally pick a model in the Model box (Auto is the default) and edit the negative prompt.
5. Press Generate or Enter. A progress bar shows the model being loaded if it changed, then the step count and, if applicable, the upscale stage.
6. The image appears in the gallery. The Upscale bar appears under it, and the details panel shows timing, settings, GPU telemetry and the exact prompt sent to the model.
7. Both files are already saved in outputs\. Clear resets the prompt and results.



Figure 4.1: "a lighthouse on a rocky cliff at golden hour, cinematic, 16:9, 4k, seed 42" rendered at 1344x768 and upscaled to 3840x2194.

4.3 Choosing a model

- Say nothing: Auto keeps Z-Image Turbo (loaded at start) or FLUX. When an SDXL-family model is loaded it switches to the specialist for a style word if that model is on disk (photorealistic to Juggernaut, anime to Animagine, oil painting or watercolor to Playground, digital art, comic or cyberpunk to DreamShaper).
- Name it in the prompt: `model juggernaut`, `model: flux`, `checkpoint=realvis`, `using dreamshaper`, `with playground`.
- Pick it in the Model box. A prompt directive always wins over the box.
- The Models accordion under the prompt lists every model, its default steps and guidance, whether it is on disk, and what it is best for.
- Steps and guidance follow the model. Quality words still scale them: `quick` halves the steps, `highly detailed` adds 35 %, `strict` and `creative` move guidance by 30 %.

4.4 Using LoRAs and your own checkpoints

- Copy an SDXL LoRA file to models\loras\, for example pixel-style.safetensors, and write `<lora:pixel-style:0.8>` anywhere in the prompt. Several tags can be stacked; the weight is optional and defaults to 1.
- Copy an SDXL checkpoint (a single .safetensors file, as distributed on Civitai) to models\checkpoints\ and restart the app. It appears in the Model box and can be named in the prompt by its file name. Files with turbo, lightning or hyper in the name get 6-step defaults.
- Two LoRAs are included: pixel-art-xl and papercut. Tick them in the LoRAs box or tag them in the prompt. SDXL LoRAs apply to SDXL-family models; a LoRA that cannot be found or does not fit the loaded model is reported in the notes under the details report rather than failing the generation.

4.5 Producing 4K and 8K output

From the prompt: include 4k, 8k, 2k or an exact size such as 3840x2160. The parser sets an upscale target and raises steps by a third for a cleaner base. Generation runs at the native bucket for the aspect ratio, then the upscaler runs one x4 pass for 4K or two passes for 8K, and an exact size is centre-cropped.

After generation: the Upscale bar under the gallery offers 2K, 4K and 8K. Click an image to upscale only that one, or leave nothing selected to upscale the whole batch. The gallery switches to the upscaled files and the history rows gain the upscale columns. The native PNG is always kept.

Recommended practice: iterate at native size with a fast model until the composition is right, then upscale the chosen image from the bar. That costs 3 to 9 seconds instead of a regeneration.



Figure 4.2: The same 300x225 region of the lighthouse cliff. Left: native pixels enlarged. Right: Real-ESRGAN output. The upscaler reconstructs rock texture rather than blurring.

4.6 Reviewing past generations

1. Open the History tab. The top line shows totals: number of images, GPU seconds, average and extreme timings, VRAM, hottest temperature and peak power.
2. Type part of a prompt in the search box and press Enter to filter, or press Refresh to reload. The table now has a model column.
3. Click a row. The image (upscaled if one exists) and its full details appear below the table.
4. For scripted access, run `db.py`, `db.py 12` or `db.py stats` with the virtual environment's Python.

4.7 Watching the machine: the Admin tab

1. Open the Admin tab. It refreshes on open and shows nine tiles and six sections; see section 9 for every field.
2. Press Refresh for a new sample, or tick Auto-refresh every 5 s while a generation runs to watch VRAM, temperature and power move.
3. Tick "Check internet reachability" to add a TCP connect test to Hugging Face, GitHub and PyPI; it is off by default because an offline machine would wait 2 s per host.

4.8 Batch generation from the command line

```
.\.venv\Scripts\python.exe generate.py --list-models
.\.venv\Scripts\python.exe generate.py "a red sports car" --model juggernaut --seed 42 --count 4
.\.venv\Scripts\python.exe generate.py "cat astronaut" --model dreamshaper --lora pixel-style:0.7
.\.venv\Scripts\python.exe generate.py --help
```

The script loads the model per run, so it is best for batches. It writes to outputs\ but does not record to the database or upscale.

4.9 Upscaling an existing image

```
.\.venv\Scripts\python.exe upscaler.py outputs\some_image.png 3840
```

The result is written next to the input with an _up3840 suffix. For images made in the app, prefer the Upscale bar so the history row is updated too.

4.10 Enabling FLUX.1

1. Accept the licence at https://huggingface.co/black-forest-labs/FLUX.1-schnell while logged in to Hugging Face.
2. Run `.\.venv\Scripts\huggingface-cli login` and paste a read token.
3. Say `model flux` in a prompt. The first use downloads about 33 GB; on a 24 GB card the transformer is quantised to 4-bit and the T5 encoder to 8-bit at load time, which takes a few minutes.
4. FLUX ignores the negative prompt, guidance words and LoRAs. Four steps are enough.

4.11 Running in Docker

1. Install the prerequisites from section 11 (on Windows: WSL 2 and Docker Desktop).
2. In the project folder run `docker compose up --build`. The first build installs PyTorch and the dependencies (several GB).
3. Open http://localhost:7860. Images land in outputs\, the database is generations.db, and downloaded models persist in the hf-cache volume.
4. Stop with Ctrl+C or `docker compose down`. The volume survives; `docker compose down -v` deletes the cached models.

4.12 Stopping, restarting and backup

- Close the console window, or press Ctrl+C in it, to stop the server. Loaded models are released.
- Restart with run-app.bat. Startup is about 30 to 45 seconds because the model is reloaded from disk.
- If port 7860 is reported busy, a previous instance is still running; end it from Task Manager (python.exe) or set GRADIO_SERVER_PORT.
- Back up the outputs\ folder, generations.db and models\ together. Built-in models can be re-downloaded and the virtual environment rebuilt from requirements.txt.

4.13 Development workflow

1. Edit a module. Self-tests: `python prompt_parser.py` prints how sample prompts are interpreted, `python models.py` lists the registry and what is on disk, `python sysinfo.py` dumps the Admin data as JSON, `python db.py stats` exercises the schema.
2. Compile-check with `python -m py_compile app.py models.py prompt_parser.py sysinfo.py`.
3. Restart the app. For automated checks, gradio_client can call /generate, /upscale, /refresh_history and /admin_refresh, which is how the increments in this project were verified.
4. Rebuild this documentation with `python docs-src\build_pdf.py`.
5. Commit and push: the repository is https://github.com/Fzhacking/ai-image-gen.

5. Module Reference

5.1 app.py (886 lines)

Entry point of the web application.

Name	Kind	Description
STARTUP_MODEL, OUTPUT_DIR, DEFAULT_NEGATIVE, UPSCALE_CHOICES, HELP	constants	Model loaded at start (zimage), outputs path, default negative prompt, the 2K/4K/8K choices, and the prompt help table.
PINK, YELLOW, GRADIENT, STATUS, CSS	constants	Brand colour ramps, the accent gradient, the four reserved status colours with icons, and the stylesheet for header, tiles and meters.
build_theme()	function	A Gradio Base theme with the Frazile AI palette applied to light and dark mode alike; only variables the installed Gradio accepts are set.
header_html(static_gpu)	function	The branded header with GPU, CUDA, torch and diffusers pills.
models_markdown()	function	The Models accordion: every model, defaults, on-disk state, plus LoRA and FLUX instructions.
MANAGER, UPSCALER, STATIC_GPU, MODEL_CHOICES	globals	The ModelManager (with sdxl loaded), the Upscaler, static GPU facts, and the dropdown choices, created once at import.
pick_model(settings, choice)	function	Resolves the model: prompt directive, then dropdown, then style-based auto choice limited to downloaded models. Returns (spec, note).
preview(prompt, choice)	function	The live "Will generate" line including the model and notes.
details_markdown(rec)	function	Renders one database record as the Markdown report shown under the gallery and in History.
upscale_record(rec, target_px, exact_size)	function	Upscales a record's native PNG, saves it, returns the eight upscale fields.
generate(prompt, negative, model_choice, progress)	function	Runs the full pipeline for all requested images. Returns gallery paths, report, records state, selection, Upscale bar visibility, hint and loaded line.
clear_all(), upscale_hint(), on_select(), upscale()	functions	Clear button, the hint under the Upscale bar, gallery selection tracking, and the Upscale bar handler that updates rows with db.update().
history_table(), history_stats(), refresh_history(), select_history()	functions	History tab, unchanged in behaviour but with a model column.
tile(), _kv(), _throttle_text(), gpu_energy_estimate(), admin_refresh(check_internet)	functions	Admin panel rendering: HTML stat tiles with meters and status, key-value tables, decoded throttle bits, the GPU energy estimate, and the collector that returns all seven outputs.

The UI is a gr.Blocks with fill_width=True and three tabs. Events: prompt change and model change to preview; Generate click and Enter to generate; Clear; gallery select to on_select; Upscale click to upscale; History tab select, Refresh and search to refresh_history; row select to select_history; Admin tab select, Refresh and a 5 s gr.Timer to admin_refresh, with the auto-refresh checkbox toggling the timer. launch() reads GRADIO_SERVER_NAME, GRADIO_SERVER_PORT and FRAZILE_NO_BROWSER and passes the theme and CSS.

5.2 models.py (436 lines)

Name	Kind	Description
ModelSpec	dataclass	key, name, source (HF repo or path), family (sdxl or flux), steps, guidance, scheduler, variant, negative, aliases, styles, description, download_gb, local.
BUILTIN	constant	The eight built-in specs with their defaults and recommended samplers.
local_checkpoints(), all_models(), model_words()	functions	Scan models/checkpoints/, the full list, and every key and alias for the parser.

Name	Kind	Description
resolve(name)	function	Finds a model by key, alias, name, repo id or path; raises KeyError if unknown.
model_for_style(style), is_downloaded(spec), display_name(source)	functions	Auto-selection lookup, on-disk check against the HF cache, and source-to-name mapping for the UI.
find_lora(name), available_loras()	functions	Case- and punctuation-insensitive lookup in models/loras/.
a1111_to_compel(prompt)	function	Rewrites (word:1.3) into compel's (word)1.3.
ModelManager	class	Holds one pipeline. load() dispatches to the SDXL, Z-Image or FLUX loader and reports progress; unload() frees VRAM; apply_loras() loads once and activates with weights; encode() returns weighted embeddings via compel with chunk padding, falling back to plain text; generation_kwargs() adds FLUX-specific arguments.

Running the module directly lists every model and whether it is downloaded.

5.3 prompt_parser.py (315 lines)

Name	Kind	Description
Settings	dataclass	prompt, width, height, steps, guidance, count, seed, style, upscale_to, exact_size, notes, model, loras, steps_explicit, guidance_explicit, quality, fidelity. Methods: summary(), upscaled_size(), apply_defaults(base_steps, base_guidance).
RATIO_BUCKETS, ORIENTATION_WORDS, STYLE_SUFFIXES, UPSCALE_TARGETS	constants	As in v1.0: eleven SDXL buckets, orientation regexes, twelve style presets, upscale words.
parse(text)	function	Returns a Settings. Order: LoRA tags, model directive, explicit resolution, ratio, orientation, upscale words, steps, guidance, count, seed, style, tidy-up, then apply_defaults with the SDXL baseline.

Quality and fidelity words no longer set fixed numbers; they are recorded and applied relative to the chosen model's defaults. Running the module directly prints the interpretation of twelve sample prompts.

5.4 sysinfo.py (358 lines)

Name	Kind	Description
human_bytes(), human_duration()	functions	Formatting helpers used by the Admin panel.
cpu()	function	Processor name (registry on Windows, /proc/cpuinfo on Linux), cores, frequency, usage per core, load averages, process and thread counts (cached 30 s), boot time, uptime.
memory(), disks()	functions	RAM and swap; partitions with usage, disk I/O totals, and the sizes of outputs/, models/, the HF cache and the database (cached 60 s).
network(), internet()	functions	Hostname, primary IP, totals, per-second rates from the previous sample, errors and drops, connections and listening ports, every interface; optional TCP reachability of three hosts.
gpu(), power()	functions	gpu_info static and live data plus PCIe link, VBIOS, max clocks, power limits, throttle reasons and GPU processes from nvidia-smi; power draw against limit, Windows power plan, battery.
software(), app(), everything()	functions	Library and driver versions and the Docker flag; the app process; one call that collects all of the above.

5.5 upscaler.py (109 lines)

Name	Kind	Description
MODEL_PATH, TARGETS, TILE, OVERLAP	constants	models/RealESRGAN_x4plus.pth; 2k 2560, 4k 3840, uhd 3840, 8k 7680; 512 and 32 pixels.
Upscaler(device, model_path)	class	Loads the model with spandrel, switches it to half precision on CUDA, exposes name and scale.
Upscaler.upscale_once(img)	method	One x4 pass over padded tiles under inference_mode.
Upscaler.upscale_to(img, long_side)	method	Repeats passes until the long side reaches the target, resizing before a second pass, then Lanczos-resamples to exactly long_side. Returns (image, info).

5.6 gpu_info.py (97 lines)

Name	Kind	Description
snapshot()	function	Runs nvidia-smi once and returns temperature_c, power_w, power_limit_w, gpu_util_pct, mem_util_pct, mem_used_mb, mem_total_mb, sm_clock_mhz, mem_clock_mhz, fan_pct, driver_version, pstate. Returns {} if nvidia-smi fails.
static_info()	function	gpu_name, vram_total_gb, compute_capability, multiprocessors, cuda_version, cudnn_version, torch_version, diffusers_version, python_version, os.
torch_memory()	function	vram_allocated_gb, vram_reserved_gb, vram_peak_gb from the torch allocator.

5.7 db.py (188 lines)

Name	Kind	Description
DB_PATH, SCHEMA, NEW_COLUMNS	constants	generations.db next to the module; CREATE TABLE plus index; columns added after the first release.
connect()	function	Opens a connection with Row factory, applies schema and migrations.
insert(record)	function	Inserts known keys as columns; unknown keys are packed into extra_json. Returns the row id.
update(row_id, fields)	function	New in v2.0: sets known columns on one row; used by the Upscale bar.
fetch(limit, search), get(row_id), stats()	functions	Newest-first list with LIKE search; one record; aggregates.

5.8 generate.py (121 lines)

Standalone command-line generator. Arguments: prompt, --negative, --steps, --guidance, --width, --height, --seed, --count, --model (key, repo id or path), --lora NAME[:WEIGHT] (repeatable), --out, --list-models. Uses the same ModelManager, so weighting, samplers and LoRAs behave exactly as in the app. It does not write to the database.

5.9 Supporting files

File	Purpose
requirements.txt	Python dependencies with the note on installing the CUDA PyTorch build first
run-app.bat	Double-click launcher that runs app.py inside the virtual environment
README.md	User-facing quick start, model table, prompt reference, Admin and Docker sections
Dockerfile, docker-compose.yml, .dockerignore	Container image and service definition (section 11)
models/RealESRGAN_x4plus.pth	Upscaler weights
models/checkpoints/, models/loras/	Drop folders for user SDXL checkpoints and LoRAs

File	Purpose
docs/	This documentation set as PDFs
docs-src/build_pdf.py, docs-src/images/	Source of this documentation and its figures

6. Models and Prompt Enhancement

6.1 Built-in models

Every model is described by a ModelSpec in models.py: where it comes from, its family, its default steps and guidance, the sampler it runs with, the style words that auto-select it and a one-line description.

Say	Model	Family	Steps	CFG	Sampler	Best for
model zimage	Z-Image Turbo	zimage	8	0	flow matching (shipped)	default model: photoreal, text rendering and strong prompt following in 8 steps
model sdxl	SDXL base 1.0	sdxl	30	7	DPM++ 2M Karras	general purpose baseline
model juggernaut	Juggernaut XL v9	sdxl	30	5	DPM++ 2M Karras	photoreal people, products and scenes; the best all-round upgrade
model realvis	RealVisXL V5.0	sdxl	30	4	DPM++ SDE Karras	photoreal with strong skin, fabric and lighting detail
model dreamshaper	DreamShaper XL v2 Turbo	sdxl	6	2	DPM++ SDE Karras	fast illustration, fantasy and stylised art
model playground	Playground v2.5	sdxl	40	3	EDM (shipped)	highly aesthetic, painterly-real composition and colour
model animagine	Animagine XL 3.1	sdxl	28	6	Euler a	anime and manga characters
model qwen	Qwen-Image + Lightning	qwen	8	1	flow matching, exponential shift	20B model in 4-bit: best text rendering and layout, posters and signage
model flux	FLUX.1 schnell	flux	4	0	flow matching (shipped)	best prompt understanding: text, counting, complex scenes

Aliases also work: z-image, zit, jugg, realistic, turbo, aesthetic, anime-xl, qwen-image, schnell. Playground keeps its own EDM scheduler because its weights were trained with it; Z-Image, Qwen-Image and FLUX use flow-matching schedulers.



Figure 6.1: The same prompt ("close-up portrait of an old fisherman with weathered skin, golden hour light, photorealistic, seed 11") rendered by the six SDXL-family models at 832x1216 with their default settings.

6.2 Automatic selection

When the Model box is on Auto and the prompt names no model, the loaded model is kept while it is Z-Image Turbo or FLUX, because both handle every style themselves. When an SDXL-family model is loaded the style preset decides: photorealistic and cinematic choose Juggernaut, anime chooses Animagine, oil painting and watercolor choose Playground, digital art, comic and cyberpunk choose DreamShaper. The switch happens only if that model is already on disk; otherwise the loaded model is kept and the notes say which model to request. With no style word the loaded model is reused, so consecutive prompts never pay a swap unless they ask for one.

6.3 Steps and guidance follow the model

Prompt says	SDXL (30 / 7)	DreamShaper Turbo (6 / 2)	FLUX (4 / 0)
nothing	30 steps, cfg 7	6 steps, cfg 2	4 steps
quick	15, 7	3, 2	2
highly detailed or 4k	40, 7	8, 2	5
strict	30, 9.1	6, 2.6	4 (note: no guidance)

Prompt says	SDXL (30 / 7)	DreamShaper Turbo (6 / 2)	FLUX (4 / 0)
creative	30, 4.9	6, 1.4	4 (note: no guidance)
25 steps cfg 3	25, 3	25, 3	25, 3

6.4 Prompt weighting and long prompts

- `(glowing sword:1.3)` or `(glowing sword)++` emphasises; `(word:0.7)` or `(word)--` de-emphasises. Automatic1111-style syntax is converted to compel's.
- Prompts longer than 77 tokens are encoded in chunks instead of being cut off. Both positive and negative embeddings are padded to the same length with the empty-prompt embedding.
- Applies to every SDXL-family model, including local checkpoints. FLUX receives the plain text; its T5 encoder reads 256 tokens natively.
- If compel ever fails on a prompt, the pipeline's own encoder is used and a line is printed; generation never fails because of weighting.

6.5 LoRAs

Two starter SDXL LoRAs are set up by `python models.py fetch-loras` (they are not kept in git): **Pixel Art XL** (nerijs/pixel-art-xl, 163 MB, trigger word "pixel art") and **Papercut** (TheLastBen/Papercut_SDXL, 325 MB, trigger word "papercut"). Tick them in the LoRAs box under the negative prompt, where they apply at weight 0.8, or write the tag in the prompt with any weight.

- Files in models/loras/ are matched by name, ignoring case and punctuation: `<lora:Pixel_Art_XL:0.9>` finds pixel-art-xl.safetensors.
- LoRAs trained for SDXL apply to every SDXL-family model. Z-Image and FLUX accept LoRAs trained for them; an SDXL LoRA requested on those models is reported as incompatible in the notes and skipped.
- Each LoRA is loaded into the resident pipeline once and then activated or deactivated per request with its weight; swapping models reloads them on demand.
- LoRAs patch the UNet and the text encoders, so they interact correctly with weighting.
- Weights are clamped to the range -2 to 2.

6.6 Local checkpoints

Any single-file SDXL checkpoint in models/checkpoints/ is loaded with diffusers' from_single_file. The key is the file name in lower-case kebab form, the description shows its size, and turbo, lightning or hyper in the name selects 6 steps, guidance 2 and the SDE sampler. The folder is scanned at start-up.

6.7 Z-Image Turbo, the default model

Z-Image Turbo (Tongyi-MAI, Apache 2.0) is a 6-billion-parameter single-stream flow transformer with a Qwen3 text encoder, distilled to run in 8 steps without classifier-free guidance. It renders photoreal scenes and legible text, follows long prompts closely, and understands English and Chinese. The app loads it at start-up and uses it whenever the prompt names no model and no style word points at a specialist. On a 24 GB card the loader enables per-model CPU offload so the text encoder leaves the GPU after encoding and the 12 GB transformer has room for activations. Like FLUX it has no negative prompt and no guidance scale, and LoRAs are not applied to it; prompt weighting is also not available because its text encoder is not CLIP.

6.8 Qwen-Image on 24 GB

Qwen-Image (Alibaba, Apache 2.0) is a 20-billion-parameter MMDiT with a Qwen2.5-VL-7B text encoder and the strongest open-weight text rendering and layout control. Its bf16 weights are 41 GB, so on a 24 GB card the loader quantises the transformer to 4-bit NF4 shard by shard straight onto the GPU on first use and saves the result under models/cache/; the text encoder comes pre-quantised from the diffusers NF4 repository. The

Lightning 8-step LoRA from lightx2v is attached as a permanent adapter with the exponential time-shift scheduler it was distilled for, which replaces the stock 50 steps with classifier-free guidance by 8 steps without it. User LoRAs stack on top of Lightning. The model has no negative prompt in this configuration and the guidance word scaling is ignored.

6.9 FLUX.1 schnell on 24 GB

FLUX.1 is a 12-billion-parameter flow-matching transformer. Its bf16 weights alone are 24 GB, so on a card under 40 GB the loader quantises the transformer to 4-bit NF4 and the T5 text encoder to 8-bit with bitsandbytes, then enables per-model CPU offload. Expect a few minutes of load time and about 15 to 25 s per 1024x1024 image. The repository is gated: the loader raises a clear error with the login instructions if the terms have not been accepted. This path is implemented but has not been exercised on the development machine because no Hugging Face token was available.

7. Prompt Language Reference

Everything about a generation is expressed in the prompt. Matching is case-insensitive. Model and LoRA tags and numeric directives are removed before the prompt reaches the model; descriptive words stay in because they also guide the image.

7.1 Model and LoRA

Write	Effect
model zimage, model juggernaut, model: flux, checkpoint=realvis	Use that model (key, alias, name, repo id or local file name)
using realvis, with dreamshaper, via flux, on playground	Same, for known model words only
<lora:name:0.8>, <lora:name>	Activate models/loras/name.safetensors at that weight (default 1)
(word:1.3), (word)++, (word)--, (word:0.7)	Emphasis weighting, kept in the prompt and applied by compel

7.2 Size and shape

Write	Effect	Generated at
16:9	Widescreen	1344x768
9:16	Vertical widescreen	768x1344
4:3 / 3:4	Classic	1152x896 / 896x1152
3:2 / 2:3	Photo	1216x832 / 832x1216
1:1	Square	1024x1024
21:9 / 9:21	Ultrawide	1536x640 / 640x1536
5:4 / 4:5	Print	1088x896 / 896x1088
1920x1080 (any WxH)	Same ratio, scaled to about 1 megapixel; if larger than native, the output is upscaled and cropped to exactly that size	depends
portrait, tall, vertical, poster, book cover	2:3	832x1216
landscape, wide, horizontal	3:2	1216x832
square, icon, avatar, profile picture	1:1	1024x1024
wallpaper, desktop wallpaper, widescreen, banner, youtube thumbnail	16:9	1344x768
phone wallpaper, mobile wallpaper, instagram story, tiktok	9:16	768x1344
ultrawide, cinemascope	21:9	1536x640

Precedence: an explicit WxH beats an explicit ratio, which beats orientation words. Among orientation words the first match in the table order wins, so "portrait of a king, landscape" is tall.

7.3 Output resolution (upscaling)

Write	Long side after upscaling	Passes
2k	2560 px	1
4k or uhd	3840 px	1
8k	7680 px	2
3840x2160 (any size above native)	exactly that size, centre-cropped	1 or 2

Any of these also raises steps by about a third unless steps are given explicitly. The same targets are offered by the Upscale bar after generation.

7.4 Quality and fidelity

Write	Effect
hd, ultra detailed, highly detailed, masterpiece, best quality, high quality	steps x 1.35 (SDXL: 40)
quick, draft, fast, rough, preview	steps x 0.5 (SDXL: 15)
25 steps, steps 25, steps=25	exactly that (1 to 100)
strict, exact, precise, literal	guidance x 1.3 (SDXL: 9.1); ignored by FLUX
creative, loose, dreamy, abstract, surreal	guidance x 0.7 (SDXL: 4.9); ignored by FLUX
cfg 8, guidance 8	exactly that (0 to 20)

Defaults come from the model: 30 steps and guidance 7 for SDXL base, see section 6.1 for the others.

7.5 Style presets

The first matching style adds descriptors that are not already present, and may auto-select a model (section 6.2).

Write	Descriptors added	Auto model
photorealistic, photo, photograph, dslr	photorealistic, 8k, DSLR photo, sharp focus, natural lighting	Juggernaut
anime, manga	anime style, vibrant colors, clean lines, studio quality	Animagine
cinematic, film still, movie scene	cinematic lighting, dramatic, film still, depth of field, color graded	Juggernaut
watercolor	watercolor painting, soft edges, paper texture, pastel tones	Playground
oil painting, oil on canvas	oil painting, visible brush strokes, canvas texture, classical	Playground
3d render, 3d, octane, blender, unreal	3d render, octane render, volumetric lighting, highly detailed	-
pixel art, 8-bit, 16-bit	pixel art, retro game sprite, crisp pixels	-
digital art, concept art, digital painting	digital painting, concept art, trending on artstation, highly detailed	DreamShaper
sketch, pencil sketch, line art, ink drawing	pencil sketch, line art, monochrome, hand drawn	-
comic, cartoon	comic book style, bold outlines, flat colors, halftone	DreamShaper
cyberpunk	cyberpunk, neon lights, rain, futuristic city, high contrast	DreamShaper
low poly	low poly, geometric, isometric, minimal shading	-

The photorealistic preset contains the word "8k" as a quality descriptor; it does not trigger 8K upscaling because upscale words are detected on the original prompt before the suffix is added.

7.6 Batch and reproducibility

Write	Effect
3 images, 4 variations, 2 versions, x4	1 to 4 images, seeds N, N+1, N+2 ...
seed 42, seed: 42, seed=42	fixed seed; omitted means random

7.7 Worked examples

Prompt	Interpreted as
a cozy cabin in a snowy forest, 16:9, 4k	1344x768, upscale to 3840x2194, 40 steps, guidance 7, loaded model

Prompt	Interpreted as
close-up portrait of an old fisherman, photorealistic	832x1216, 30 steps, guidance 5, Juggernaut XL v9 (auto)
a knight in silver armour, (glowing sword:1.3), model realvis, quick	1024x1024, 15 steps, guidance 4, RealVisXL V5.0, sword emphasised
neon city street at night using flux, 16:9	1344x768, 4 steps, guidance 0, FLUX.1 schnell
cat astronaut <lora:pixel-style:0.7> with dreamshaper	1024x1024, 6 steps, guidance 2, DreamShaper, LoRA pixel-style x0.7
anime girl with umbrella, tall, strict, 2 variations	832x1216, 28 steps, guidance 7.8, Animagine (auto), 2 images
oil painting of a ship in a storm 21:9 steps 45 cfg 8	1536x640, 45 steps, guidance 8, Playground (auto)

8. Database

8.1 Overview

File: generations.db in the project folder. Engine: SQLite 3 through Python's standard library. One table, generations, one row per image produced by the web app. An index on created_at supports newest-first listing. The schema is created on first connection; columns added in later increments are listed in NEW_COLUMNS and added with ALTER TABLE when absent. Since v2.0 rows can be amended after creation: the Upscale bar writes the upscale columns onto an existing row with db.update().

8.2 Table: generations

Identity and files

Column	Type	Meaning
id	INTEGER PK	Auto-increment row id, shown as "#N" in the UI
created_at	TEXT	ISO-8601 local timestamp, seconds precision
image_path	TEXT	Absolute path of the native PNG
file_size_kb	REAL	Size of the native PNG

Prompt

Column	Type	Meaning
user_prompt	TEXT	Exactly what was typed, including model and LoRA tags
final_prompt	TEXT	What was sent to the model after directive removal and style suffixes
negative_prompt	TEXT	Negative prompt used; empty for FLUX
parser_notes	TEXT	Parser decisions, auto-model note, LoRA notes and the sampler, joined with " - "

Settings

Column	Type	Meaning
model_id	TEXT	Model source: Hugging Face repo id or local checkpoint path
width, height	INTEGER	Native generation size
steps	INTEGER	Denoising steps actually used (after model defaults)
guidance	REAL	Guidance scale actually used
seed	INTEGER	Seed used for this image
style	TEXT	Name of the matched style preset, or NULL
batch_index, batch_size	INTEGER	Position within a multi-image request

Timing

Column	Type	Meaning
inference_seconds	REAL	Pipeline call, from start to CUDA synchronise
save_seconds	REAL	PNG encode and write of the native image
total_seconds	REAL	inference + save (excludes upscaling)
steps_per_second	REAL	steps / inference_seconds

GPU and software

Column	Type	Meaning
gpu_name	TEXT	From torch device properties
vram_total_gb	REAL	Card memory
vram_peak_gb	REAL	Peak torch allocation during this image, including an in-prompt upscale
vram_reserved_gb	REAL	Allocator reservation after the image
compute_capability	TEXT	e.g. "8.6"
driver_version	TEXT	From nvidia-smi
cuda_version, torch_version, diffusers_version	TEXT	Library versions

Telemetry (from nvidia-smi)

Column	Type	Meaning
temp_before_c, temp_after_c	REAL	GPU temperature before the pipeline call and right after it
power_before_w, power_after_w	REAL	Board power draw at the same two moments
power_limit_w	REAL	Configured power limit
gpu_util_after_pct	REAL	GPU utilisation at the after snapshot
sm_clock_mhz, mem_clock_mhz	REAL	Clock speeds at the after snapshot
fan_after_pct	REAL	Fan speed
pstate	TEXT	Performance state, P0 to P8

Upscaling (written at generation or later by the Upscale bar)

Column	Type	Meaning
upscale_target	INTEGER	Requested long side in pixels
upscaled_width, upscaled_height	INTEGER	Final size after resample and crop
upscaled_path	TEXT	Absolute path of the upscaled PNG
upscaled_size_kb	REAL	Size of that file
upscale_seconds	REAL	Time of the upscaling stage
upscale_passes	INTEGER	Number of x4 passes (1 or 2)
upscaler	TEXT	Architecture and weight file name

Catch-all

Column	Type	Meaning
extra_json	TEXT	JSON of any record keys that have no column, for forward compatibility

8.3 Example queries

Average inference time by model:

```
SELECT model_id, COUNT(*) AS n, ROUND(AVG(inference_seconds), 2) AS avg_s,
       ROUND(AVG(steps_per_second), 2) AS steps_s
FROM generations GROUP BY model_id ORDER BY n DESC;
```

Hottest runs:

```
SELECT id, temp_after_c, power_after_w, user_prompt
FROM generations ORDER BY temp_after_c DESC LIMIT 5;
```

Every 4K or larger image with its file:

```
SELECT id, upscaled_width, upscaled_height, upscale_seconds, upscaled_path
FROM generations WHERE upscale_target >= 3840;
```

Estimated GPU energy of all generations, as the Admin panel computes it:

```
SELECT ROUND(SUM(inference_seconds * (power_before_w + power_after_w) / 2) / 3600, 1) AS wh
FROM generations;
```

8.4 Access from Python

```
import db
rows = db.fetch(limit=10, search="cabin")
one = db.get(4)
db.update(4, {"upscaled_width": 3840})
totals = db.stats()
```

9. Admin Panel

The Admin tab is a live view of the machine running the studio. Every number is sampled on demand by sysinfo.py; nothing is stored. A refresh costs about 0.7 s (3 s the first time, when folder sizes are computed). The tab refreshes when opened, on the Refresh button, and every 5 s when auto-refresh is ticked.

9.1 Tiles

Tile	Value	Meter or state	Source
GPU load	utilisation %, SM clock, P-state	meter	nvidia-smi
VRAM used	GiB used of total; torch allocation	meter of used %	nvidia-smi, torch
GPU temperature	°C and fan %	state: OK under 70, Warm under 83, Hot under 90, Critical above	nvidia-smi
GPU power	W of limit, % of limit	meter	nvidia-smi
CPU	usage %, threads, frequency	per-core bar strip	psutil
Memory	GB used of total	meter	psutil
System disk	GB free, % used	meter	psutil
Network	download and upload rate, established connections	-	psutil, two samples
App uptime	uptime, PID, resident memory	-	psutil

Meters use the pink accent and switch to the amber warning colour above 85 % and the red critical colour above 95 %. The temperature state always carries an icon and a word, never a colour alone.

9.2 Sections

Section	Fields
GPU	Card, VRAM, compute capability and SM count; driver, CUDA, cuDNN; current and maximum SM and memory clocks; GPU and memory-bus utilisation; VRAM from nvidia-smi and from torch (allocated, reserved, session peak); PCIe generation and width; throttle reasons decoded from the nvidia-smi bitmask (idle, power cap, thermal slowdown and so on); VBIOS and PCI bus; loaded model and sampler; processes on the GPU.
CPU	Processor name, architecture, physical and logical cores, current and maximum frequency, total and per-core usage, 1/5/15-minute load averages, process and thread counts, boot time and uptime.
Memory and storage	RAM used, available and total; swap or page file; disk reads and writes since boot; outputs folder size and PNG count; models folder; Hugging Face cache size and path; database size; every partition with file system, total, used, free and percentage.
Network	Hostname and primary IP; current throughput (rates need two samples); bytes and packets since boot; errors and drops; established connections and listening ports; every interface with state, IPv4, MAC, link speed, sent and received; optional reachability of huggingface.co, github.com and pypi.org in milliseconds.
Power	GPU draw against the configured limit with percentage; default and maximum limits; performance state and fan; the active Windows power plan; battery if present; estimated GPU energy of all recorded generations in Wh.
Software and app	OS, Python, torch, CUDA, cuDNN, diffusers, transformers, gradio versions; whether the process runs in Docker; app PID, start time, uptime, resident memory and threads; working directory; generation totals from the database.

9.3 Reading the numbers

- Idle readings on the RTX 3090 are about 45 W, P8, SM clock 210 MHz, PCIe gen 1 x1 and throttle reason "GPU idle". Under load expect P0, 1.7 to 1.9 GHz, PCIe gen 3 or 4 x16 and 350 to 390 W.
- nvidia-smi reports allocator reservation as used VRAM; the torch allocated figure is the live tensor total.

- Process names on the GPU may show "Insufficient Permissions" for processes of other users or the system.
- The network rate tiles show 0 on the first sample after start; every later refresh shows the rate since the previous one.
- Inside Docker the CPU, memory and network sections describe the container, while GPU and power come from the host driver.

10. Performance

All figures were measured on the development machine (RTX 3090, 24 GB, driver 610.88, torch 2.14.1 with CUDA 12.6) and taken from generations.db and the v2.0 model sweep.

10.1 Generation by model (832x1216, prompt weighting on for SDXL)

Model	Steps	CFG	Sampler	Inference	Peak VRAM
Z-Image Turbo	8	0	Flow-match Euler	8.6 s (9.8 s at 1344x768)	18.1 GB
SDXL base 1.0	30	7	DPM++ 2M Karras	8.0 s	13.1 GB
Juggernaut XL v9	30	5	DPM++ 2M Karras	8.0 s	13.1 GB
RealVisXL V5.0	30	4	DPM++ SDE Karras	8.0 s	13.2 GB
DreamShaper XL v2 Turbo	6	2	DPM++ SDE Karras	2.0 s	13.2 GB
Playground v2.5	40	3	EDM DPM-Solver	10.5 s	13.2 GB
Animagine XL 3.1	28	6	Euler a	7.3 s	13.2 GB

Every SDXL-family model runs at about 3.7 to 3.8 steps per second at this size. Inference time is therefore proportional to the step count, which is why the turbo model's 6 steps finish in 2 seconds. Z-Image Turbo runs at about 1.1 s per step but needs only 8, so a finished image takes 9 to 10 s with better prompt adherence and legible text. Loading an SDXL-family model from the local cache takes 26 to 27 s; freeing it returns resident VRAM to 0.01 GB. Measurements taken while another process holds GPU memory are not valid: with the web app resident alongside the test, Z-Image took 78 to 128 s per image because Windows paged VRAM.

10.2 Generation (v1.0 measurements, 1024x1024 and 16:9, SDXL base)

Size	Steps	Inference	Steps/s	Peak VRAM	Temp after	Power after
1024x1024	15	4.7 to 4.9 s	3.1 to 3.2	10.5 GB	55 to 56 C	377 to 379 W
1024x1024	30	9.7 s	3.1	10.5 GB	-	-
1344x768	40	12.8 to 13.5 s	3.0 to 3.1	10.4 GB	52 to 54 C	349 to 357 W
768x1344	40	11.2 to 11.4 s	3.5 to 3.6	10.5 GB	55 to 57 C	382 to 385 W

10.3 Upscaling

From	To	Passes	Time	Output file
1344x768	3840x2194 (4K)	1	3.4 s	9.9 MB
768x1344	2194x3840 (4K)	1	2.9 s	7.9 MB
768x1344	1463x2560 (2K, Upscale bar)	1	3.6 s	3.5 MB
768x1344	4388x7680 (8K)	2	8.4 to 8.5 s	30.5 MB

10.4 Memory

Stage	VRAM
Idle with one SDXL-family model loaded	6.6 GB allocated
Idle with Z-Image Turbo loaded (bf16 transformer + 8-bit text encoder)	15.8 GB allocated
Peak during Z-Image generation at 1344x768	18.1 GB
Peak during 1 MP generation, plain prompt	10.4 to 10.5 GB
Peak during 832x1216 generation with compel embeddings	13.1 to 13.2 GB

Stage	VRAM
After unload	0.01 GB
Upscaler weights	67 MB (half precision)
Peak during tiled upscale	no measurable increase over the generation peak

10.5 Startup and switching

Step	Time
First ever start (downloads 6.7 GB)	bandwidth dependent; the five fine-tunes downloaded in 76 to 184 s each on the development connection
SDXL load from local cache at start	34 to 59 s (disk cache dependent)
Z-Image Turbo first load (fp32 to bf16 conversion plus writing the 12 GB cache)	about 250 s
Z-Image Turbo load from models/cache/	about 130 s (8-bit quantisation of the text encoder dominates)
Model switch while running (SDXL family)	26 to 27 s
Upscaler load	under 1 s
Admin panel refresh	about 3 s first, 0.7 s after

10.6 Tuning levers

- Model: Z-Image Turbo gives a finished photoreal image in 9 s; DreamShaper Turbo gives a rougher draft in 2 s. For SDXL-specific looks or LoRAs, regenerate the chosen seed with Juggernaut or RealVis.
- Steps: the dominant cost for every model. `quick` halves them; `highly detailed` adds a third.
- Count: images in a batch are generated sequentially, so time scales linearly.
- Upscale later: the Upscale bar costs 3 to 9 s per image and avoids regenerating.
- Cards with less VRAM: generate.py enables CPU offload below 12 GB, which roughly doubles inference time but keeps the pipeline working.

11. Deployment and Docker

11.1 Image

The Dockerfile builds on `nvidia/cuda:12.6.3-cudnn-runtime-ubuntu24.04`, installs Python 3.12, PyTorch and torchvision for CUDA 12.6, then requirements.txt, and copies the Python sources. The Hugging Face cache is pointed at /cache/huggingface, Gradio is told to bind 0.0.0.0:7860, and FRAZILE_IN_DOCKER=1 stops the app from trying to open a browser. A health check polls the root page every 30 s after a 3-minute start grace.

11.2 Compose service

Setting	Value	Why
ports	7860:7860	The studio on http://localhost:7860
volumes	./outputs, ./models, ./generations.db, hf-cache	Images, your checkpoints and LoRAs, the history database, and the downloaded models all live outside the container
environment	HF_TOKEN	Optional; lets the gated FLUX repository download
deploy.resources.reservations.devices	driver nvidia, count all, capabilities gpu	GPU passthrough
shm_size	2 GB	Shared memory for data loaders
restart	unless-stopped	Comes back after a reboot of the host

11.3 Host requirements

Host	Needs
Windows 11	WSL 2 (`wsl --install`, reboot), Docker Desktop with the WSL 2 backend, the normal NVIDIA Windows driver. GPU support is built into Docker Desktop; no extra toolkit.
Linux	Docker Engine, the NVIDIA driver, nvidia-container-toolkit configured as a Docker runtime.

The development machine has neither WSL nor Docker installed, so the image has been written and reviewed but not built or run there. The first build on any machine downloads several GB of Python packages and should be run on a fast connection.

11.4 Commands

```
docker compose build           # build the image
docker compose up              # run in the foreground
docker compose up -d           # run detached
docker compose logs -f         # follow the app log
docker compose down            # stop; keeps the hf-cache volume
docker compose down -v         # stop and delete cached models
```

11.5 Differences inside the container

- Paths in the database are container paths (/app/outputs/...). Mount the same folders when moving between native and container use, or the History tab shows "not found" for the other side's files.
- The Admin panel's CPU, memory and network sections describe the container's view; GPU and power still come from the host driver through nvidia-smi.
- On Linux hosts create an empty generations.db before the first `up`, otherwise Docker creates a directory in its place.
- There is still no authentication. Publishing port 7860 beyond localhost exposes the studio to the network.

12. Operations and Troubleshooting

12.1 Starting and stopping

Action	How
Start	Double-click run-app.bat, or `.\.venv\Scripts\python.exe app.py` in the project folder; or `docker compose up`
Confirm ready	Console shows "SDXL base 1.0 ready in N s", "Upscaler ready", "Running on local URL: http://127.0.0.1:7860"
Stop	Ctrl+C in the console, or close the window; `docker compose down` for the container
Change port or address	Set GRADIO_SERVER_PORT or GRADIO_SERVER_NAME before starting; FRAZILE_NO_BROWSER=1 suppresses the browser tab
Expose on the LAN	GRADIO_SERVER_NAME=0.0.0.0. There is no authentication, so only do this on a trusted network

12.2 Logs

The app prints one line per loaded model, per generated image and per upscale, for example:

```
Loading Juggernaut XL v9 ...
Juggernaut XL v9 ready in 26.3s  (scheduler: DPM++ 2M Karras)
#12 juggernaut saved 20261022-101040_seed1746058488.png  832x1216  8.0s  peak 13.1 GB  56.0 C
#12 upscaled -> 1463x2560 in 3.56s
```

To keep a log file, redirect: `.\.venv\Scripts\python.exe -u app.py > app.log 2>&1`. Inside Docker use `docker compose logs -f`.

12.3 Known issues encountered during development and their fixes

Symptom	Cause	Fix applied
AttributeError: 'StableDiffusionXLPipeline' object has no attribute 'enable_vae_slicing'	diffusers 0.41 removed the pipeline-level helper	Call pipe.vae.enable_slicing() instead
Parser ignored "seed 42" and misread "21:9 steps 45" as 9 steps	Directive removal used offsets from a stale lowercase copy of the prompt	Rewrote removal to search and cut the live prompt string each time
Regex word boundaries silently vanished	A patch script wrote "\b" through a normal Python string, turning it into a backspace	Patches now use raw strings or the file is rewritten whole
compel: 'EmbeddingsProviderMulti' object has no attribute 'empty_z'	compel's pad_conditioning_tensors_to_same_length does not support SDXL's two encoders, so a long prompt with a short negative failed	models.py pads the shorter embedding with chunks of the empty-prompt embedding
Theme build failed: Base.set() got an unexpected keyword argument 'color_accent_dark'	Not every Gradio theme variable has a dark-mode twin	build_theme() filters keys against the signature of Base.set()
Admin refresh took 3 s every time	Walking the 39 GB Hugging Face cache and every process for thread counts	Folder sizes cached 60 s, thread count cached 30 s; refresh is 0.7 s
Segmentation fault while generating with Z-Image Turbo	enable_model_cpu_offload() with the 23 GB fp32 transformer on a 16 GB RAM machine	Transformer loaded shard by shard onto the GPU in bf16 with device_map; text encoder in 8-bit; bf16 copy cached under models/cache/
Z-Image generation took 78 to 128 s	The web app was still holding 7 GB of VRAM during the benchmark, so Windows paged GPU memory	Stop other GPU processes before benchmarking; with the GPU free the same images take 8.6 to 9.8 s
"strict" reported higher guidance for FLUX	Fidelity words were noted before the model was known	apply_defaults() notes that the model has no guidance scale

Symptom	Cause	Fix applied
Space before a comma after a removed tag ("cat , quick")	Tag removal left the separator behind	Tidy step also collapses whitespace before commas
pip could not write pymupdf.exe in C:\Python312\Scripts	Permission problem in the system Python's Scripts folder	Not a project issue; the venv is unaffected. The documentation tooling used the system Python only for one-off edits
UnicodeEncodeError when printing reports from scripts	Windows console code page cp1252 cannot print the arrow and dot characters	Set PYTHONIOENCODING=utf-8 when printing reports; the browser is unaffected
torchvision installed as the CPU build	pip pulled torchvision from PyPI before the CUDA index was consulted	Reinstalled from the cu126 index; requirements.txt documents the order

12.4 Troubleshooting guide

Problem	Check	Resolution
"CUDA GPU not available" at startup	`python -c "import torch; print(torch.version.cuda)"` prints None	Reinstall torch from the cu126 index inside the virtual environment
Port 7860 already in use	Task Manager shows a python.exe from a previous run	End that process, or set GRADIO_SERVER_PORT
"Unknown model 'x'" in the preview line	Typo, or a checkpoint added without restarting	Use a key from the Models accordion; restart after adding files to models/checkpoints/
Generation waits 30 s before starting	The progress bar says "Loading <model>"	Normal for a model switch. Keep the Model box on one model, or Auto, for consecutive prompts
First use of a model waits minutes	Progress says "downloading ~6.5 GB"	Wait; check the HF cache folder growth
"FLUX.1 [schnell] is gated"	No Hugging Face token stored	Accept the terms on the model page and run huggingface-cli login (section 4.10)
LoRA has no effect and the notes say "not found"	File name does not match the tag	Compare with the LoRA list in the Models accordion; matching ignores case and punctuation
Out of memory during generation	Other GPU programs running; or a card under 12 GB; or Z-Image or FLUX on under 24 GB	Close other GPU apps; use generate.py which offloads to CPU on small cards; Z-Image and FLUX need 24 GB
Z-Image is very slow (over a minute per image)	Another process holds several GB of VRAM; nvidia-smi shows memory near 24 GB	Close it; Z-Image needs about 18 GB free
Telemetry fields show None	nvidia-smi not on PATH	Reinstall the NVIDIA driver, or add C:\Windows\System32 to PATH
Upscaler weights missing	FileNotFoundError naming models/RealESRGAN_x4plus.pth	Download from the Real-ESRGAN v0.1.0 release into models/
Upscale bar does nothing	No generation in this browser session	Generate first; the bar works on the records of the last generation
History tab empty	New database, or generations.db moved	Generate an image; or restore the file next to db.py
Admin panel shows 0 B/s network	First sample	Refresh again; rates need two samples
Browser slow to show an 8K result	30 MB PNG transfer	Expected; open the file from outputs/ directly
docker compose fails with "could not select device driver"	GPU support not enabled in Docker	Install WSL 2 and Docker Desktop (Windows) or nvidia-container-toolkit (Linux)

12.5 Maintenance

- Outputs grow by 1.5 MB per native image and 3 to 31 MB per upscaled image. Prune outputs/ periodically; the History tab will show "not found" for deleted files. The Admin panel shows the folder size.
- Each built-in model is 6.5 GB in the Hugging Face cache. Delete a models--* folder there to reclaim space; it downloads again on next use.
- Back up outputs/, generations.db and models/ together so paths stay valid.
- To reset history, stop the app and delete generations.db; it is recreated empty on next start.
- To update dependencies, upgrade inside the virtual environment and rerun the parser self-test, models.py, sysinfo.py and one generation per family. Pin versions in requirements.txt if a breaking change is found.

12.6 Security notes

- The server listens on localhost only by default and has no authentication. The Docker image binds all interfaces inside the container; the compose file publishes the port on the host, so treat that like a LAN exposure.
- No telemetry or prompts are sent anywhere. Network access is limited to model downloads from Hugging Face and the optional reachability check in the Admin panel, which only opens a TCP connection.
- The Admin panel reveals hostnames, IP addresses, MAC addresses, listening ports and process names. Anyone who can reach the page can read them.
- Prompts are stored verbatim in generations.db. Treat the file with the same care as the images.

13. Change Log and Roadmap

13.1 Change log

Increment 1: command-line generator (21 Oct 2026)

- Created the project folder, virtual environment, and requirements.txt; installed torch 2.14.1+cu126, diffusers 0.41, transformers, accelerate.
- Wrote generate.py with prompt, negative, steps, guidance, size, seed, count and model arguments. First image: 1024x1024 in 9.7 s, peak VRAM 10.5 GB.

Increment 2: web interface

- Added app.py using Gradio 6.30 with a gallery and an info box; model loaded once at startup and kept resident. Added run-app.bat and README.

Increment 3: prompt-driven controls

- Removed style, size, steps and guidance controls. Added prompt_parser.py, the live "Will generate" line and a collapsible help table.

Increment 4: telemetry and history

- Added gpu_info.py and db.py with a 45-column SQLite table; per-image timing and GPU sampling; the History tab.

Increment 5: real 4K and 8K output

- Added upscaler.py (Real-ESRGAN x4plus, tiled, multi-pass); parser upscale words; eight upscale columns; 4K in 3.4 s, 8K in 8.5 s.

Documentation 1.0 and repository (22 Oct 2026, 09:17)

- Documentation set published as PDFs under Frazile AI ownership. Project pushed to github.com/Fzhacking/ai-image-gen (private).

Increment 6: model registry (22 Oct 2026, 10:09)

- Added models.py: SDXL base, Juggernaut XL v9, RealVisXL V5.0, DreamShaper XL v2 Turbo, Playground v2.5, Animagine XL 3.1 and FLUX.1 schnell; local checkpoints; LoRAs; compel weighting; per-model samplers and defaults; ModelManager with verified VRAM release.
- Parser: model and LoRA directives, explicit-flag tracking, apply_defaults(). App: Model dropdown with Auto, model in details and history. CLI: --model, --lora, --list-models.
- Installed peft 0.21, compel 2.5, bitsandbytes 0.50, sentencepiece, protobuf. Downloaded the five fine-tunes (32.5 GB).
- Fixed: compel padding for SDXL dual encoders; comma tidy-up; FLUX fidelity note.
- Measured: every SDXL-family model at 832x1216 in 2.0 to 10.5 s, peak 13.2 GB, swap 26 s.

Increment 7: Frazile AI branding and Upscale bar (22 Oct 2026, 10:27)

- Dark studio theme with pink-to-yellow accents, applied to both colour modes; branded header; Inter and JetBrains Mono.
- Upscale bar after generation: 2K / 4K / 8K, selected image or whole batch, records updated in place with the new db.update().

Increment 8: full width, Admin panel, Docker (22 Oct 2026, 10:50)

- Full-width layout, version pills, Clear button, "Loaded now" line.

- Admin tab with nine tiles and six sections; sysinfo.py collector on psutil and nvidia-smi; 5 s auto-refresh; optional reachability check; caching so a refresh costs 0.7 s.
- Dockerfile, docker-compose.yml and .dockerignore; app reads bind address, port and browser flag from the environment.

Increment 9: Z-Image Turbo as the default model (22 Oct 2026)

- Added the zimage family to models.py with a bf16 loader and CPU offload under 32 GB; Z-Image Turbo (Tongyi-MAI, 6B, 8 steps, Apache 2.0) is the model loaded at start-up.
- The repository ships the transformer in fp32 (23 GB); the first load converts it to bf16 and saves a 12 GB copy under models/cache/ so later loads take about about 130 s. The Qwen3 text encoder is loaded in 8-bit; everything stays on the GPU (15.8 GB resident, 18.1 GB peak).
- Measured: 1344x768 in 9.8 s and 832x1216 in 8.6 s at 8 steps; the FRAZILE AI neon-sign test rendered the text correctly.
- Auto no longer switches away from Z-Image or FLUX; style-based specialist selection applies only while an SDXL-family model is loaded.
- Fixed: CPU offload of Z-Image crashed the process on the 16 GB machine (fp32 shards plus offload exceeded system RAM); replaced by shard-by-shard loading straight onto the GPU.

Documentation 2.0 (22 Oct 2026)

- This document set, rebuilt from docs-src/build_pdf.py with three new sections (Models, Admin panel, Deployment) and a model comparison figure.

13.2 Roadmap

Priority	Item	Notes
High	Image-to-image and inpainting	Same pipeline family; lets a chosen image be edited rather than regenerated; the ModelManager already shares components
High	Character consistency with IP-Adapter or reference images	Keeps a subject's identity across prompts; the usual next step after LoRAs
High	Test the Docker image on a host with WSL 2 and Docker Desktop	Written and reviewed but unbuilt on the development machine
Medium	FLUX.1 dev and ControlNet	dev is gated and non-commercial; ControlNet gives pose and depth control
Medium	SDXL refiner or hires-fix pass	Adds generated detail at 2x before Real-ESRGAN
Medium	Export history to CSV and chart timings per model	pandas is already present
Medium	Thumbnail cache for the History grid	Avoids loading full PNGs when browsing
Low	Authentication	Gradio supports basic auth; needed before exposing beyond localhost or publishing the container port
Low	Cloud deployment profile	The Docker image is the starting point for RunPod or a Hugging Face Space

Appendix

A. Project file tree

```
ai-image-gen/
  app.py                  web application (Gradio), theme, Admin panel
  models.py               model registry, ModelManager, LoRAs, compel
  prompt_parser.py        prompt -> settings
  sysinfo.py              system telemetry for the Admin panel
  upscaler.py             Real-ESRGAN tiled upscaling
  gpu_info.py             GPU facts and live telemetry
  db.py                   SQLite store and CLI viewer
  generate.py             command-line generator
  requirements.txt
  run-app.bat
  README.md
  Dockerfile, docker-compose.yml, .dockerignore
  generations.db          created on first generation
  models/
    RealESRGAN_x4plus.pth
    cache/                bf16 copies of converted models (auto-built, 12 GB for Z-Image)
    checkpoints/          your SDXL .safetensors files
    loras/                your SDXL LoRA files
  outputs/                generated PNG files
  docs/                   this documentation set as PDFs
  docs-src/
    build_pdf.py          source of the documentation
    images/               figures
  .venv/                  virtual environment (not for backup)
```

B. requirements.txt

```
# Install PyTorch with CUDA first (RTX 3090 / any recent NVIDIA card):
#   pip install torch torchvision --index-url https://download.pytorch.org/whl/cu126
# Then:
#   pip install -r requirements.txt
diffusers>=0.31
transformers>=4.45
accelerate>=1.0
safetensors
huggingface_hub
gradio>=5.0
pandas
psutil>=5.9           # Admin panel: CPU, memory, disk and network stats
spandrel>=0.4          # Real-ESRGAN upscaler
peft>=0.13             # LoRA loading
compel>=2.0            # prompt weighting (word:1.3) and long prompts for SDXL models
sentencepiece          # T5 tokenizer for FLUX
protobuf               # T5 tokenizer for FLUX
bitsandbytes>=0.45     # 4-bit FLUX on 24 GB cards
```

C. Command cheat sheet

All commands run from the project folder in PowerShell with `.\.venv\Scripts\python.exe`.

Task	Command
Start the app	app.py
List models	generate.py --list-models
Generate from CLI	generate.py "prompt" --model juggernaut --seed 42 --count 2
Generate with a LoRA	generate.py "prompt" --model dreamshaper --lora name:0.8
Upscale a file	upscaler.py outputs\x.png 3840
List history / one record / statistics	db.py | db.py 4 | db.py stats

Task	Command
Parser self-test	prompt_parser.py
Model registry status	models.py
Fetch the starter LoRAs	models.py fetch-loras
GPU telemetry dump	gpu_info.py
Full system telemetry as JSON	sysinfo.py
Log in to Hugging Face (for FLUX)	.\.venv\Scripts\huggingface-cli login
Rebuild the PDFs	docs-src\build_pdf.py
Run in Docker	docker compose up --build
Check CUDA	-c "import torch; print(torch.cuda.is_available())"

D. Programmatic access to the running app

```
from gradio_client import Client
c = Client("http://127.0.0.1:7860")
images, report, records, sel, bar, hint, loaded = c.predict(
    "a lighthouse, 16:9, seed 42", "", "juggernaut", api_name="/generate")
table, stats = c.predict("lighthouse", 50, api_name="/refresh_history")
tiles, gpu, cpu, mem, net, power, soft = c.predict(False, api_name="/admin_refresh")
```

E. Licences

Component	Licence
Stable Diffusion XL base 1.0	CreativeML Open RAIL++-M
Juggernaut XL v9, RealVisXL V5.0, DreamShaper XL v2 Turbo	CreativeML Open RAIL++-M
Playground v2.5	Playground Community License
Animagine XL 3.1	Fair AI Public License 1.0-SD
Z-Image Turbo, Qwen-Image, Qwen-Image-Lightning, FLUX.1 schnell	Apache 2.0
Real-ESRGAN weights and code	BSD 3-Clause
diffusers, transformers, accelerate, peft, Gradio	Apache 2.0
compel, spandrel, bitsandbytes	MIT
psutil, PyTorch	BSD-style
Project source code	Owned by Frazile AI

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